MIL-C-12999B

14 December 1967

SUPERSEDING
MIL-C-12999A
7 March 1963

MILITARY SPECIFICATION

CHEST, HYMN BOOK

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type and size hymn book chest (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.
TT-E-529	-	Enamel, Alkyd, Semi-Gloss.
TT-I-558	-	Ink, Marking Stencil, Opaque, for Nonporous Surfaces (Metals, Glass, etc.).
TT-P-636	-	Primer Coating, Alkyd, Wood and Ferrous Metal.
MMM-A-181	-	Adhesive, Room-Temperature and Intermediate Temperature Setting Resin (Phenol, Resorcinol, and Melamine Base).
PPP-B-636	-	Box, Fiberboard.

MILITARY

MIL-F-10336	-	Fiber Sheet, Vulcanized.

STANDARDS

FEDERAL

Fed. Std. No. 595	-	Colors.

FSC 9925

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U. S. Military Property.

DRAWING

ARMY NATICK LABORATORIES

14-1-144 - Chest, Hymn Book; Assembly and Details.

(Miniature copy of Drawing 14-1-144, identified as figure 1, is attached for information purposes only.)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U. S. DEPARTMENT OF COMMERCE

Commercial Standard CS 35-61 - Hardwood Plywood.

Product Standard PS 1-66 - Softwood Plywood, Construction and Industrial

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Preproduction sample.- When specified (see 6.2), before production is commenced, a sample chest covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.4. The approval of the preproduction sample chest shall not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction chest shall be manufactured in the same facility to be used for the manufacture of the contract items.

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3.2 Materials.- Materials not definitely specified shall be of the quality normally used for the purpose.

3.2.1 Plywood.- Plywood used in the fabrication of the chest shall be 3/8 inch thick. Each of the panels for the chest shall be one piece of plywood conforming to either of the following:

(a) Hardwood, type I, fully waterproof bond, sound grade, (2) face and back or better, polish sanded two sides, any species, conforming to Commercial Standard CS 35-61.

(b) Softwood, exterior type, group 1, grade B-B, sanded two sides, conforming to Product Standard PS 1-66, or better.

3.2.2 Fiber sheet, vulcanized.- Vulcanized fiber sheet shall conform to MIL-F-10336. Unless otherwise specified herein, the thickness shall be not less than 0.018 nor more than 0.022 inch.

3.2.3 Adhesive.- Room temperature and intermediate temperature setting adhesive shall conform to type I or II of MMM-A-181, as applicable. High temperature setting adhesive shall conform to the requirements for type II of MMM-A-181, except that the requirements for shear strength by compression loading and for intermediate temperature cure cycle need not apply.

3.2.4 Primer.- Primer shall conform to TT-P-636.

3.2.5 Enamel.- The enamel shall conform to class A of TT-E-529. The color shall be olive drab conforming to Color No. X24087 of Fed. Std. No. 595.

3.2.6 Stencil ink.- The stencil ink shall conform to TT-I-558. The color shall be black.

3.2.7 Hardware.- The hardware shall be heavy duty chest hardware, fabricated from steel in accordance with the manufacturer's commercial practice, except the minimum metal thicknesses of steel and steel wire diameters shall be as follows (commercial tolerances shall apply):

Hinges	-	0.070 inch
Valance clamps	-	0.056 inch
Corner and valance	-	0.058 inch
Corner bottom	-	0.058 inch
Drawbolt (body)	-	0.058 inch
Drawbolt (loop)	-	0.187 inch

The ends of drawbolt loops shall be closed and welded.

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3.2.8 Rivets, trunk tacks and burrs.- All rivets shall be oval head. Solid and tubular rivets may be used interchangeably. Trunk tacks and split rivets may be used interchangeably for attaching the fiber binding. Steel rivet burrs shall be of the size recommended by the rivet industry and shall be fabricated from steel strip 0.055 to 0.070 inch thick. Trunk tacks shall be oval or shot head with square shanks. The length of split rivets and trunk tacks shall be sufficient to provide a minimum of 1/8 inch clinch. The length of tubular or solid rivets shall be sufficient to provide a minimum of 1/16 inch for peening or curling.

3.3 Design and construction.- The chest shall conform to Drawing 14-1-144 and to the requirements specified herein. The chest panels shall be fabricated of the plywood specified in 3.2.1. The plywood shall be covered on both sides with vulcanized fiber sheet specified in 3.2.2, using the adhesive specified in 3.2.3. The chest panels shall be assembled at right angles, glued with the adhesive specified in 3.2.3, and nailed so that the body, lid, hardware, and tongue and groove binding on the chest shall mate. All hardware shall be assembled with rivets and burrs.

3.3.1 Fiber covered plywood panels.- Plywood panels, prior to covering with fiber sheet, shall be cleaned of dust and foreign matter. Each side shall be covered by a single sheet of fiber. The covered panel shall be free of defects such as bubble, gouge, tear, hole, wave, ripple or seam. The adhesive bond of fiber to plywood shall show a wood fiber failure of not less than 95 percent prior to priming, enameling or treating, when tested as specified in 4.4.1.

3.3.2 Fiber edge binding fillers and quarter circles.- Fiber used for edge bindings, fillers and quarter circles shall be made of the fiber specified in 3.2.2, except the thickness shall be 0.048 inch. Fiber strips used for edge bindings shall be set tightly to the edge and tacked or riveted. Each fiber binding shall be made from a single piece of material.

3.3.3 Metal bindings.- Metal tongue and groove bindings shall be formed with true right angle bends, properly notched to provide tight mitered joints at corners, and shall set tightly against the surface of the chest when riveted in position. Bindings shall be formed either in one unit or from four pieces having tight mitered joints.

3.3.4 Application of adhesives.- Application of adhesives, assembly periods, pressing time, pressure and curing temperature shall be in accordance with the adhesive manufacturer's recommendations.

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3.3.5 Driving of nails, trunk tacks and rivets.- All solid and tubular rivets shall be driven with heads flush against hardware, burrs, or fiber with the ends securely headed or set. Holes for solid or tubular rivets shall be drilled through fiber and plywood and shall be no larger than the diameter of the rivet. Rivet holes in hardware shall be drilled or punched not more than 0.015 inch larger than the rivet diameter. All split rivets, trunk tacks and nails shall be thoroughly driven with heads flush against hardware, wood or fiber. Rivets and tacks shall be clinched to provide a smooth finish on interior surfaces.

3.4 Finish.-

3.4.1 Metal surface preparation.- Prior to priming, the surfaces of steel components (except trunk tacks, rivets and burrs) shall be given a cleaning and surface preparation conforming to type I or II of TT-C-490.

3.4.2 Priming.-

3.4.2.1 Fiber covered surfaces.- Prior to attachment of hardware, a coat of primer specified in 3.2.4 shall be applied to all inside and outside fiber covered surfaces of the chest. The prime coat shall level out to produce a dry film without runs, wrinkles, streaks or areas of no film, and shall be free from grit or coarse particles. Forced drying is permitted, provided that a surface temperature of 140 F. is not exceeded and the relative humidity of surrounding air is maintained at 30 to 50 percent.

3.4.2.2 Metal parts.- Prior to attaching to the chest, all metal parts, excluding rivets, trunk tacks and burrs, shall be given a coat of primer specified in 3.2.4. Baking of the primer is permissible and the dry film shall level out to a smooth uniform film.

3.4.3 Enameling.- After the attachment of hardware, the complete chest shall be given a full hiding coat of enamel specified in 3.2.5. All surfaces shall be dry before stacking or packing.

3.5 Marking for identification.- The chest exterior shall be marked in the locations shown on Drawing 14-1-144, using black ink specified in 3.2.6. Marking shall be in accordance with MIL-STD-130 and shall contain the following information:

U. S.
Date of manufacture
Nomenclature
Contract order number
Manufacturers' identification or trademark

3.6 Workmanship.- Metal components shall be free from sharp edges, slivers or burrs. Fiber covering shall be free from tears and scuffs. Wood shall not be split nor shall nail points protrude through wood or fiber. Enameling shall be dry, free from runs, wrinkles, streaks, areas of no film, and grit or coarse particles imbedded under finish.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection.- When a preproduction sample is required, it shall be examined for defects in table II, dimensions specified, and tested in accordance with 4.4.1.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2 In-process inspection.- Inspection shall be made during the manufacturing process for the requirements specified in table I to establish that no deviation is made from indicated requirements. Whenever nonconformance is noted, correction shall be made to affected items and process.

TABLE I. - In-process inspection

Requirement	Paragraph
Cleaning of plywood surfaces	3.3.1
Application of adhesive	3.3.4
Driving of nails, trunk tack and rivets	3.3.5
Diameter of rivet holes	3.3.5
Metal surface preparation	3.4.1
Primer application	3.4.2
Surfaces dry before stacking or packing	3.4.3

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4.3.3 Intermediate testing, prior to assembly.- The test specified in 4.4.1 shall be performed on fiber covered plywood to determine compliance with 3.3.1. The inspection lot shall be all the fiber covered plywood offered for inspection at one time. The sample unit shall be five (5) specimens, 4 inches square, each selected from a different piece and from different locations on the fiber covered plywood. The inspection level shall be S-2. Failure of any specimen shall be cause for rejection of the lot.

4.3.4 End item inspection.- The inspection lot shall consist of all completely fabricated chests offered for inspection at one time. The sample unit shall be one completely fabricated chest.

4.3.4.1 Visual examination.- Examination of the chests shall be in accordance with the classification of defects in table II. The inspection level shall be level II with an AQL of 2.5 for major defects, 6.5 for major and minor A, and 25.0 total, expressed in terms of defects per hundred units.

TABLE II. - Classification of defects

Examine	Defect	Classification Major	Minor A	Minor B
Finish (all components)	Not finished	X		
	Wrong type			X
	Not specified color			X
	Not adherent, e.g., blistered, peeled or flaked		X	
	Not smooth and uniform, e.g., run, drop streak or evidence of coat (prime coat shows through)			
	Exterior		X	
	Interior			X
	Area of no film	X		
	Foreign matter imbedded			X
Construction and workmanship (General)	Fiber covering contains bubble, gouge, tear, hole, wave, ripple or seam		X	
	Chest panels not assembled at right angles	X		

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TABLE II. - Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Construction and workmanship (General) (cont'd)	Components do not mate or function properly	X		
	Any component not securely attached		X	
Attachment of valances and bindings	Binding interferes with opening or closing of chest	X		
	Bindings consist of more than one piece, per side		X	
	Ends of binding not butted where required		X	
	Ends of binding are visible after assembly of hardware		X	
	Tongue and groove binding not formed as specified		X	
	Any binding not a tight fit to chest		X	
Rivets, burrs, and riveting:				
Rivets	Not type specified		X	
	Head of rivet or trunk tack not flush with component		X	
	Insufficient stock to permit specified curl or peen		X	
	Rivets peened or curled against fiber		X	
	Peening or curling results in sharp rough surface	X		
	One rivet or trunk tack missing, per line			X
	Two rivets or trunk tacks missing per line		X	
	More than two rivets or trunk tacks missing, per line	X		
Burr (as required)	Missing		X	
	Wrong size		X	

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TABLE II. - Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Nails	Point protrudes	X		
	One head broken off		X	
	Two heads broken off	X		
Marking	Missing, misspelled, illegible, wrong type, color or size, not applied in specified manner nor in the specified location			X

4.3.4.2 Dimensional examination.- Dimensional examination shall be made on a completed chest to determine compliance with this specification. Any dimension that is not within the specified requirements shall be classified as a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.3.5 Examination of preparation for delivery requirements.- An examination shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be in accordance with table III. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of shipping containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 2.5, expressed in terms of defects per hundred units.

TABLE III. - Examination of packaging, packing, and marking

Examine	Defect
Marking (exterior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application
Materials	Any component missing
	Any component damaged

4.3.5.1 Examination of shipping container.- When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure shall be in accordance with the appendix of the specification.

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4.4 Tests.-

4.4.1 Vulcanized fiber adhesion test.- The specimens specified in 4.3.3 shall be immersed in boiling water for three (3) hours. The specimens shall be cooled in 70 F. plus or minus 5 F. water for a period of ten (10) minutes and immediately examined, while wet, for delamination or weakness of bond at the glue line of fiber and plywood. Specimens passing the test for delamination or weakness at glue line shall be subjected, while wet, to the following test: The sharp edge of a pocketknife blade or similar instrument shall be inserted at the glue line, and the fiber covering the plywood shall be peeled off (between the thumb and knife blade). The percentage of fiber failure wood material adhering to glue line shall be calculated for compliance with 3.3.1.

5. PREPARATION FOR DELIVERY

5.1 Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A.- Each chest shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic, grade 200 of PPP-B-636. Closure shall be in accordance with method II of the appendix of PPP-B-636.

5.1.2 Level B.- Each chest shall be packed as specified in 5.1.1, except that fiberboard shall be grade 125.

5.1.3 Level C.- Chests shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Shipments shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.2 Marking.- In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129. Special handling marking requirements applicable to arrows and the word "UP" shall apply.

6. NOTES

6.1 Intended use.- The hymn book chest is intended for storage and shipment of hymn books.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When preproduction sample is required (see 3.1).
(c) Selection of applicable level of packing (see 5.1).

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6.3 Asterisks have not been used in this revision to identify changes with respect to the previous issue due to the extensiveness of changes.

Custodians:

Army - GL
Navy - PE
Air Force - 28

Preparing activity:

Army - GL

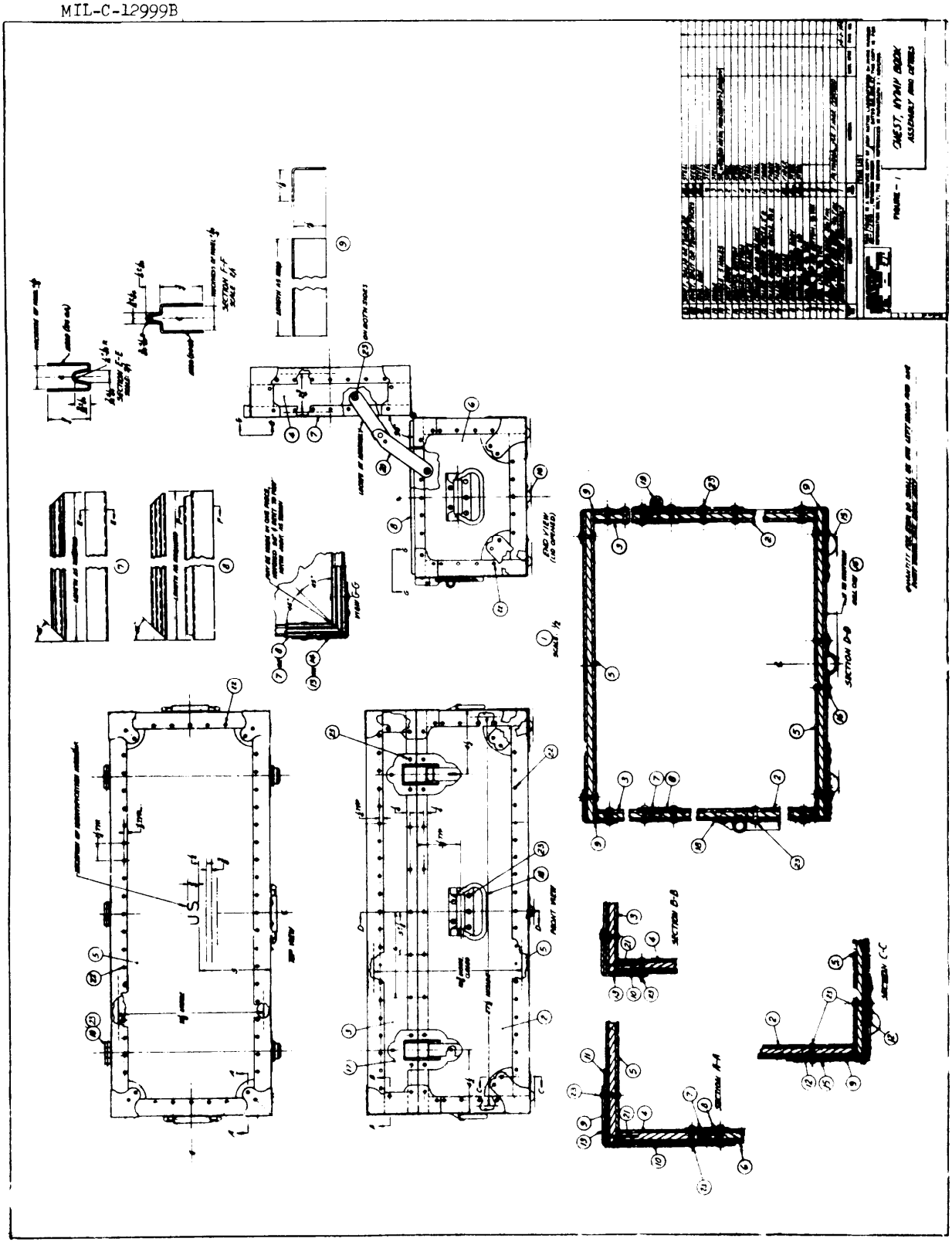
Project No. 9925-0120

Review activities:

Navy - PE
Air Force - 28
Armed Forces Chaplains Board

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Standardization Documents.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-12999B; Chest, Hymn Book

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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